JOURNAL OF BIOSCIENCE AND BIOENGINEERING
Vol. 105, No. 1, 1–11. 2008
DOI: 10.1263/jbb.105.1

© 2008, The Society for Biotechnology, Japan

REVIEW

Stoichiometric Modelling of Cell Metabolism

Francisco Llaneras[1]* and Jesús Picó[1]

Institute of Automatic Control and Industrial Informatics, Technical University of Valencia, Camino de Vera s/n, 46022 Valencia, Spain[1]

Received 26 July 2007/Accepted 25 October 2007

There are several methodologies based on representations of cell metabolism that share two characteristics: the use of a metabolic network and the assumption of pseudosteady state. These methodologies have different purposes, employ different mathematical tools and are based on different assumptions; however, they all exploit the properties of a similar mathematical description. In this article we use the term stoichiometric modelling to encompass all these methodologies and to describe them within a common framework. Although the information about reaction stoichiometry embedded in metabolic networks is highly important, the framework encompasses methodologies not limited to the use of stoichiometric information. To highlight this fact, the definition of the framework is approached from a constraint-based perspective. One of the reasons for the success of stoichiometric modelling is that it avoids the difficulties that arise in the development of kinetic models: a consequence of the lack of intracellular experimental measurements. Thus, it makes it possible to exploit the knowledge about the structure of cell metabolism, without considering the still not very well known intracellular kinetic processes. Stoichiometric models have been used to estimate the metabolic flux distribution under given circumstances in the cell at some given moment (metabolic flux analysis), to predict it on the basis of some optimality hypothesis (flux balance analysis), and as tools for the structural analysis of metabolism providing information about systemic characteristics of the cell under investigation (network-based pathway analysis).

[**Key words:** stoichiometric modelling, metabolic networks, flux balance analysis, metabolic flux analysis, elementary modes, extreme pathways, flux spectrum approach]

An observed cellular behavior cannot be explained only by considering the constitutive elements of a cell. To define all its capabilities, and also to predict its behavior, the interactions between elements need to be taken into consideration; *i.e.*, a systemic approach is needed. This confers a crucial role to networks because they embed these interactions; therefore, they are responsible for observable cellular behavior (1). Examples of networks used in biology include metabolic, regulatory and signalling networks; however the first is the best characterized one.

Conventionally, metabolism is divided in individual metabolic pathways, which are in fact a central paradigm in biology. These pathways were defined on the basis of their step-by-step discovery, but this is nowadays changing: fostered by the new reality of biology, characterized by data richness and the importance of studying cell metabolism under a system-level approach, the set of metabolic pathways involved in whole-cell metabolism is being assembled in metabolic networks (2). In most cases, these networks are highly detailed, but smaller networks can also be formulated.

In this article, we use the term stoichiometric modelling framework to encompass the methodologies based on representations of cell metabolism that share two characteristics: the use of a metabolic network and the pseudosteady state assumption (see Fig. 1). On the one hand, stoichiometric models are derived from the metabolic network of an organism under investigation. The information about reaction stoichiometry embedded in these networks is the starting point; however, the framework encompasses methodologies not limited to the use of stoichiometric information. To illustrate this point, a constraint-based perspective will be used. On the other hand, stoichiometric modelling disregards the dynamic intracellular behavior, on the basis of the assumption of the pseudosteady state for internal metabolites (3). Although this can be surprising initially, there are two main reasons for using this assumption: (i) this avoids the difficulties in developing kinetic models, a consequence of the lack of intracellular experimental measurements (4, 5), and (ii) the purely structural analysis of metabolism provides information about steady-state characteristics of a cell under

* Corresponding author. e-mail: frallaes@doctor.upv.es
phone: +34-654700098 fax: +34-963879579

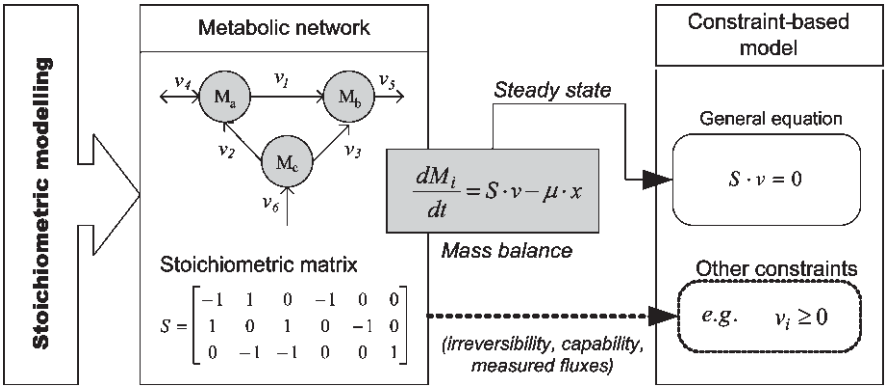

FIG. 1. Principles of stoichiometric modelling framework. The stoichiometric information embedded in a metabolic network can be represented with a matrix that translates biological knowledge into mathematical terms. Using this matrix, the mass balances for each intracellular metabolite can be mathematically represented by a set of ordinary differential equations. Then, in stoichiometric modelling, dynamic intracellular behavior is disregarded on the basis of assumption of pseudosteady state for the internal metabolites. Thereby, the mass balances can be described by a homogeneous system of linear equations, the so-called general equation. This equation constraints the flux distributions that can be achieved by the metabolic network, but it does not predict the actual distribution. To this end, additional constraints, such as irreversibility or capacity constraints, can be incorporated in order to determine what functional states, *i.e.*, flux distributions, can and cannot be achieved by a cell under certain conditions.

investigation. These characteristics may represent systemic properties of cell metabolism (*e.g.*, robustness, redundancy, existence of simplest pathways). In summary, the stoichiometric modelling framework exploits structurally detailed models, but at the cost of disregarding intracellular kinetics.

This review article is structured as follows: Firstly, the classical principles of the stoichiometric modelling framework are summarized, and the framework is reviewed from the constraint-based perspective. Then, the applications and methodologies within the framework are classified, and the most important ones are described; namely, metabolic flux analysis, flux balance analysis, and network-based pathway analysis. In the last section, we summarize the article and offer some final conclusions.

CLASSICAL PRINCIPLES OF STOICHIOMETRIC MODELLING

A metabolic network is defined by a set of compounds (nodes) and a set of reactions or fluxes that connect some compounds with others (arrows). The stoichiometric information embedded in a metabolic network with m metabolites and n reactions can be represented by a stoichiometric matrix, in which rows correspond to metabolites and columns to reactions. This matrix is of chief importance because it represents the translation of biological knowledge into mathematical terms.

Once the stoichiometric matrix has been determined, mass balances involving intracellular metabolites can be mathematically represented by a set of ordinary differential equations:

$$\frac{d\mathbf{c}}{dt} = \mathbf{S} \cdot \mathbf{v} - \mu \cdot \mathbf{c} \tag{1}$$

where $\mathbf{c}=(c_1, c_2, ..., c_m)$ is the vector of intracellular metabolites concentration, $\mathbf{v}=(v_1, v_2, ..., v_n)$ the flux vector, and μ the specific growth rate of cells. This is the dynamic mass balance equation, and it describes the evolution over time of the concentration of each metabolite, c_i. This equation implies that, in order to model the dynamic evolution of intracellular metabolites, information about stoichiometry (**S**), biomass growth (μ), and intracellular reaction fluxes (**v**) is needed.

Unfortunately, the mechanisms of intracellular reactions are complex and still not very well understood (4, 5). For this reason, in stoichiometric modelling, we disregard dynamic intracellular behavior on the basis of assumption of the pseudosteady state for internal metabolites (3). This assumption is supported by the observation that intracellular dynamics are much faster than extracellular dynamics. Therefore, it is sensible to assume that this dynamics reach the steady state instantaneously and to disregard its transient behavior. In addition, the dilution term $\mu \cdot \mathbf{c}$ is also disregarded because it is generally much smaller than the fluxes affecting the same metabolite (3). Note that the steady state assumption does not imply that the dynamic nature of the entire process is completely disregarded, because dynamic extracellular processes, such as substrates uptake and product formation, can still be considered.

Under these assumptions, the mass balances of each metabolite can be described by a homogeneous system of linear equations, the so-called general equation:

$$\mathbf{S} \cdot \mathbf{v} = 0 \tag{2}$$

This equation defines a space where every feasible flux distribution **v** lives. It restricts flux distributions that can be achieved by a metabolic network, but it does not predict the actual one. This is sensible since cells can show different behaviors depending, for example, on the availability of substrates or environmental conditions. Thus, Eq. 2 represents, or more precisely, contains, the capabilities of the metabolic network being modelled. Assuming that **S** has full row rank, there are m independent equations. As n is typically larger than m, the system is underdetermined with n-m degrees of freedom. The space defined by the solution of Eq. 2 is, by extension, said to be underdetermined.

TABLE 1. Most common types of constraint

Constraint	Type	Mathematical formulation
Systemic stoichiometry	Nonadjustable	$\mathbf{S}\cdot\mathbf{v}=0$ (the general equation)
Irreversibility of fluxes	Nonadjustable	$v>0$
Enzyme/transporters capacities	Nonadjustable	$v<v_{\max}$
Measured fluxes	Adjustable	$v=v_m$ or $v_{m,\min}<v<v_{m,\max}$
Regulatory constraints	Adjustable	Example: $v_1=0$ if $(v_2\neq 0)$
Kinetic constants	Adjustable	$v=k\cdot C_m$

C_m, Measured concentration.

CONSTRAINT-BASED PERSPECTIVE

Constraint-based modelling is based on the fact that cells are subject to governing constraints that limit their behavior (1). In principle, if all constraints operating under a given set of circumstances are known, the actual state of a metabolic network can be elucidated. However it has been suggested that most likely we will not be able to reach this state of knowledge, at least in a foreseeable future (5, 6). Nevertheless, imposing the known constraints, it can be determined which functional states, or flux distributions, can and cannot be achieved by a cell. Thus, the imposition of constraints leads to a space of feasible flux distribution rather than to the prediction of a particular solution, as it happens with the general equation (Eq. 2). In biological terms, the constraints define a space where every feasible flux distribution lives (7). Moreover, since the metabolic phenotype can be defined in terms of flux distributions, this space represents, or at least contains, all the feasible phenotypes (8).

Importantly, the general Eq. 2 is a set of stoichiometric constraints. This establishes the link between stoichiometric modelling and constraint-based modelling. The classical stoichiometric model based on the general equation can be viewed as a particular case of a constraint-based model that considers only stoichiometric constraints.

Different types of constraint Constraints can be divided in two main types: non adjustable (invariant) and adjustable (Table 1). The former are time-invariant restrictions of possible cell behavior, whereas the latter depend on environmental conditions, may change through evolution, and vary from one individual cell to another (9). Examples of non adjustable constraints are those imposed by thermodynamics (*e.g.*, irreversibility of fluxes) and enzyme or transport capacities (*e.g.*, maximum flux values). Enzyme kinetics, regulation, and experimental measurements are examples of adjustable constraints. To study the invariant properties of a network, only invariant constraints can be used, because they are always satisfied (*i.e.*, they limit the capabilities of a cell). Conversely, when adjustable constraints are used, the elucidated cell state is valid under a particular set of circumstances in which constraints exist.

Space of feasible flux distributions The general equation, Eq. 2, provides a set of stoichiometric constraints that link some fluxes with others, and restrict the space of possible flux distributions to a hyperplane, subspace of $\mathbf{R}^n$ where each axis corresponds to the flux through a reaction in a network (Fig. 2).

Afterwards, irreversible fluxes which assumed to flow in only one direction are naturally constrained to be non-negative (by convention, the natural flow direction is assigned the positive direction).

$$v_i \geq 0 \tag{3}$$

Since these restrictions are inequalities, the resulting system is beyond the scope of linear algebra. However, convex analysis shows that the space of flux distributions formed using Eqs. 2 and 3 has been converted into a convex polyhedral cone (10), as shown in Fig. 2.

Finally, maximum flux values derived from enzyme or transport capacities can be imposed.

$$\mathbf{v} < \mathbf{v}_{\max} \tag{4}$$

If this information is available for every flux in the network, the space of possible solutions is bounded (indeed, it is sufficient if each edge of the cone has a constrained flux). In mathematical terms, the convex polyhedral cone has been converted into a polytope (Fig. 2).

Equations 2–4 represent the most common non adjustable constraints. Because these constraints are time-invariant, they define a space wherein every feasible flux distribution will always live. That is, it is a model of the capabilities of the metabolism under study. This space is obtained using the general equation and by incorporating more biological knowledge, in the form of constraints, into our model. Thereby, this mathematical description becomes the core of the entire stoichiometric modelling framework.

It is also possible to incorporate non adjustable constraints to further restrict the space of possible flux distributions or even to predict the actual flux distribution. For example,

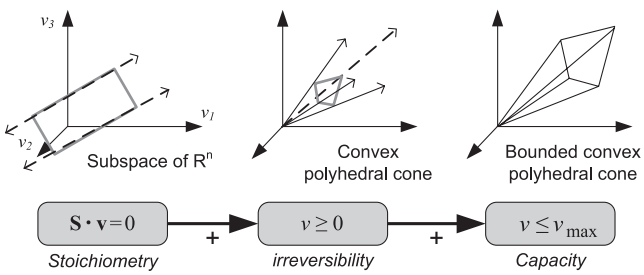

FIG. 2. Consideration of nonadjustable constraints to define the space of feasible steady-state flux distributions. The stoichiometric constraints restrict the space of possible flux distributions to a hyperplane, subspace of $\mathbf{R}^n$. If the irreversible fluxes are constrained to be non-negative, the resulting space of flux distributions is a convex polyhedral cone. Maximum capacity constraints, derived from enzyme or transport capacities, can also be imposed, and if they are sufficient in number, the space of possible solutions becomes bounded (a polytope).

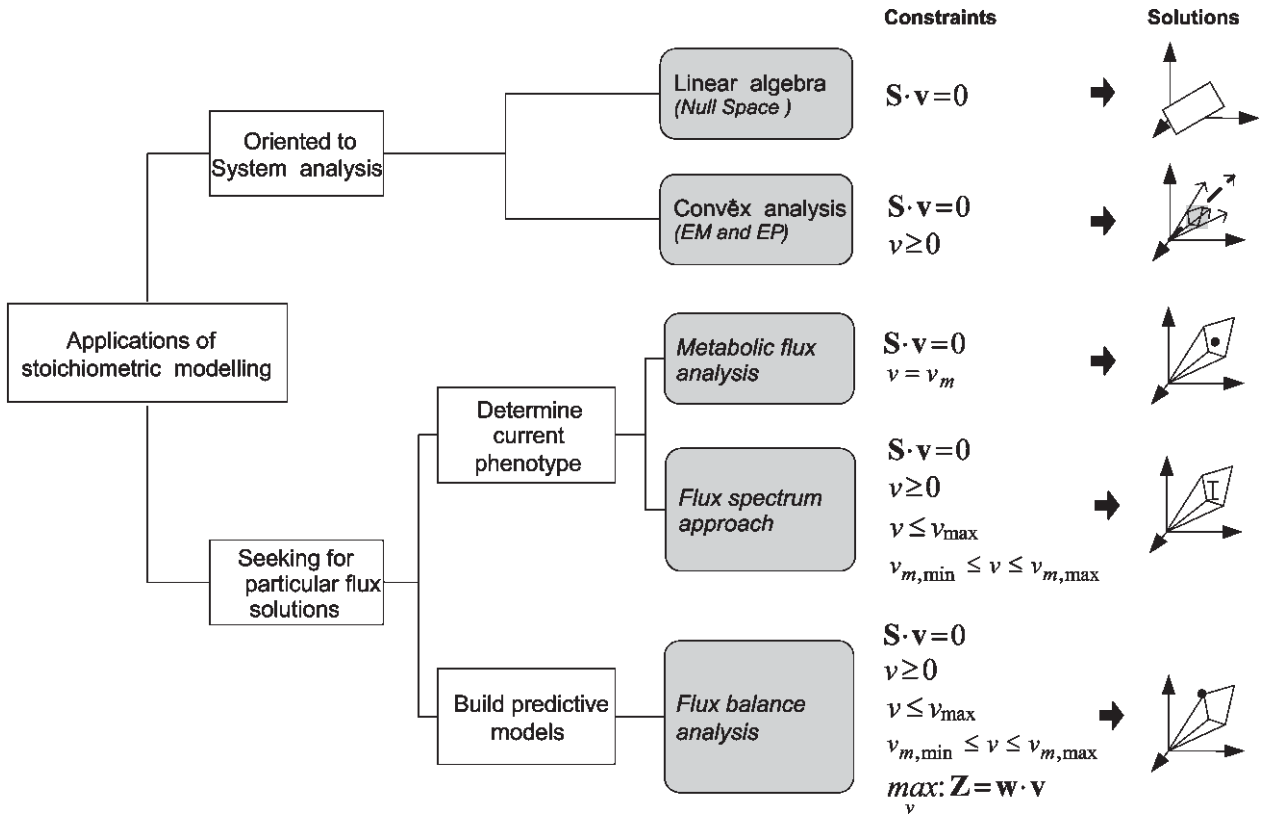

FIG. 3. Scheme of different applications and methodologies within the stoichiometric modelling framework. They can be classified in two main categories: those focused on the properties of the whole space of possible flux distributions, and those for determining particular flux solutions of this space.

regulatory constraints have been successfully imposed using Boolean logic operators (11), correlated reactions (12), and control-effective fluxes (13). As will be shown, there are methodologies that incorporate experimental flux values as adjustable constraints.

METHODOLOGIES WITHIN STOICHIOMETRIC MODELLING

There are several methodologies that exploit the mathematical description of cell metabolism which is the basis of stoichiometric modelling: the space of possible flux distributions defined by Eqs. 2–4. However, each methodology has a different purpose (*e.g.*, to analyze a network), employs a different mathematical framework (*e.g.*, linear algebra), and is based on different assumptions (*e.g.*, optimal behavior), among others. A rational classification of these methodologies can be based on their division between those focused on the properties of the entire space of possible flux distributions, and those for determining particular flux solutions in it (14) (see Fig. 3).

Applications for system analysis Certain methods are used to analyze the properties of the space of flux distributions defined by Eqs. 2–4, thereby elucidating systemic (emergent) properties of the organism under investigation.

Network-based pathway analysis based on linear algebra This approach is based on the analysis of the null space of a stoichiometric matrix, which contains all the cell steady states (flux distributions) fulfilling Eq. 2. One can gain an insight into pathway structures within a metabolic network by calculating biochemically meaningful basis vectors for the null space (15). The null space has also been used in the context of metabolic control analysis (16, 17).

Network-based pathway analysis based on convex analysis Convex analysis enables the analysis of linear systems of inequalities, thus making it possible to consider the irreversibility of fluxes using Eq. 3. Two similar approaches, elementary modes and extreme pathways analysis (2, 18), use convex analysis to generate unique convex sets of vectors that characterize all the steady state flux distributions of a metabolic network. Both are being used to elucidate systemic properties of the organism under investigation, *e.g.*, pathway length, network redundancy, or enzyme subsets.

Applications for determining a flux solution under given conditions In several methodologies within the stoichiometric modelling framework, experimental measurements are used to obtain values for certain fluxes and thus reduce, or even offset, the underdeterminacy of the space defined by Eqs. 2–4. This approach is mainly used to (i) estimate the current flux distribution using all the available measurements, or (ii) develop models capable of predicting the unique feasible flux distribution under certain conditions. This philosophy has been also used in some analytical studies. Thus, for example, the effect of measured uptake and secretion fluxes on the size of the space of possible flux

distributions was investigated by Wiback *et al.* (7).

The space of possible flux distributions can be coupled with experimental measurements of fluxes in order to completely determine the current flux distribution. Basically, every measurable flux is measured *in vivo*, and the measured values are used to offset the underdeterminacy of the space of possible flux distributions. This is the approach used with metabolic flux analysis. It provides a methodology to uniquely determine the current flux distribution using Eq. 2 and a set of measured fluxes, commonly extracellular ones (3). Metabolic flux analysis has been extensively applied in recent years, and it has been particularly successful in the fields of microbial production and animal cell culture. Recently, the flux spectrum approach has been introduced as a variant of classical metabolic flux analysis (9, 19).

A predictive model is a mathematical representation of a system that, given certain inputs, is capable of predicting the evolution of the system outputs. In our context, certain constraints can be seen as inputs to a cell, mainly substrates uptakes and other environmental conditions. A model of a cell may receive this information as input, and then predict the corresponding flux distribution. However, as we do not know all governing constraints, when the input constraints are incorporated, a unique prediction is not obtained, but a space of feasible steady state flux distributions. To determine which of these flux distributions corresponds to the actual one, flux balance analysis, which follows this approach, is carried out assuming that cell behavior is optimal with respect to a (known) objective, and the optimal flux distribution is calculated using an optimization routine (20, 21). An increasing number of organisms are being modeled using this methodology (12, 22–26), and it has been shown that flux balance analysis predictions are consistent with experimental data (12, 27).

ESTIMATION OF FLUXES: METABOLIC FLUX ANALYSIS

In metabolic flux analysis, a set of measured fluxes (mainly extracellular fluxes) is used in combination with knowledge about the cell metabolism, described by means of the general equation, to determine those fluxes that have not been measured (3). The metabolic flux analysis provides a metabolic flux map that represents the steady state at which each reaction in a network occurs (Fig. 4). This map provides information about the contribution of each reaction to the overall metabolic processes of substrate utilization and product formation.

Coupling constraints and measurements It is possible to use the constraints imposed by the measured fluxes to offset the underdeterminacy of the system defined by Eq. 2. Hence, by considering a partition between measured (subindex *m*) and unmeasured or unknown fluxes (subindex *u*), we can obtain the fundamental equation of metabolic flux analysis.

$$\mathbf{S_u} \cdot \mathbf{v_u} = -\mathbf{S_m} \cdot \mathbf{v_m} \tag{5}$$

Thereby, the system defined by Eq. 5 integrates the stoichiometric constraints given by Eq. 2 and those imposed by measured fluxes. This system can be analyzed following the

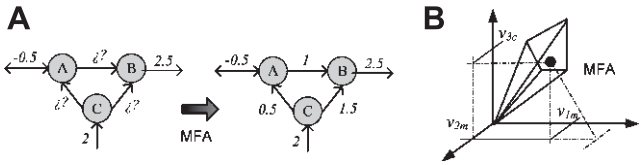

FIG. 4. Metabolic flux analysis. (A) The measured fluxes are combined with the stoichiometric constraints provided by the network to determine remaining fluxes. In this case, the three measured fluxes are sufficient to determine the three unknown fluxes. (B) The underdeterminacy of the space of feasible flux distributions is offset by incorporating certain measured fluxes. The subindex *m* denotes the measured fluxes and *c* the determined ones.

definitions given by Klamt *et al.* (28). System 5 is determined when there are sufficient linearly independent constraints to uniquely calculate all unmeasured fluxes $\mathbf{v_u}$; *i.e.*, when $rank(\mathbf{S_u})=u$ (u being the number of unmeasured fluxes). On the contrary, the system is classified as underdetermined when at least one unmeasured flux, and probably most of the fluxes, is not calculable. System 5 is redundant when some rows in $\mathbf{S_u}$ can be expressed as linear combinations of other rows; *i.e.*, when $rank(\mathbf{S_u})<m$. This can lead to an inconsistent system if the vector $\mathbf{v_m}$ contains values such that no $\mathbf{v_u}$ exists that exactly solves 5. Thereby, when the system is redundant, the inconsistency of the measurements can be checked and its importance can be estimated (3). If inconsistency occurs, it is possible to estimate its importance to prevent the use of erroneous measurements. However it must be noticed that even when they system is redundant, only the consistency of certain measured fluxes can be analyzed. These fluxes are called balanceable. The balanceable fluxes can be detected as explained in Ref. 3, and they should be adjusted (or balanced) in case the system is inconsistent (3). The value of the rest of measured fluxes has no effect on system consistency.

When the system is determined and is not redundant, the unique set of values for the unmeasured fluxes that satisfy Eq. 5 can be calculated using the inverse matrix of $\mathbf{S_u}$: $\mathbf{v_u} = -\mathbf{S_u}^{-1} \cdot \mathbf{S_m} \cdot \mathbf{v_m}$. If the system is determined but is redundant, the matrix $\mathbf{S_u}$ is not invertible, so the Penrose pseudoinverse is used instead (providing a least squares solution): $\mathbf{v_u} = -\mathbf{S_u}^{\#} \cdot \mathbf{S_m} \cdot \mathbf{v_m}$. However, if Eq. 5 is underdetermined, the classical metabolic flux analysis cannot be used.

Applications of metabolic flux analysis Metabolic flux analysis has been widely used to characterize canonical states of cells, such as exponential batch growth or steady states in the continuous mode (29, 30). In particular, animal cell cultures have received considerable attention (31–34). Recently, there has been an increasing interest on the application of metabolic flux analysis to plant cell culture (35–37). It has also been applied to study transient processes in microorganisms (38), to on-line monitoring intracellular fluxes in mammalian cells (39), to determine the physiological state of a culture (40), and to develop dynamic models (41). There are some medical applications of metabolic flux analysis, such as the generation of hypothesis for new therapeutical strategies (42), the optimization of an extracorporeal bioartificial liver device (43), and the investigation of metabolic responses of the rat liver to burn-injury-induced whole-body inflammation (44). Finally, it must be noted that met-

abolic flux analysis has been combined with isotopic labelling experiments, allowing for a more reliable estimation of fluxes (30, 45–47). More applications of metabolic flux analysis in the context of metabolic engineering are described by Shimizu (48).

Flux spectrum approach Although metabolic flux analysis has been successfully applied for quite many years, it has some limitations. The most critical one is that a considerable number of fluxes should be determined for metabolic flux analysis to provide results. Unfortunately, the measurable fluxes are insufficient in many cases, *i.e.*, the system defined by Eq. 5 is underdetermined. Moreover, only stoichiometric constraints are used, while other well-known constraints, such as reversibility constraints, are not exploited. Another difficulty is that the effect of errors in the measurements on the fluxes determined by metabolic flux analysis is not under control. This is particularly important because flux measurements are, in general, imprecise and insufficient. Although there are methods to deal with the presence of errors, they are based on redundancy, so additional measurements are needed in order to use them. Yet, this disagrees with the common lack of measurable fluxes.

To overcome some of these limitations, the flux-spectrum approach has been introduced and applied to monitoring metabolic fluxes over time (9, 19). It provides a more reliable and richer estimation of unmeasured fluxes when the system is determined. This is due to the inclusion of reversibility constraints (or other inequalities) and the consideration of the intrinsic uncertainty of experimental measurements. Moreover, it can be used to estimate unmeasured fluxes even when there is a lack of measurable species, *i.e.*, when Eq. 5 is underdetermined and therefore metabolic flux analysis cannot be used. It also offers an interesting procedure to deal with inconsistency when the system is redundant and a method to detect some inconsistencies even when the system is not redundant.

PREDICTIVE MODELLING: FLUX BALANCE ANALYSIS

The procedure for developing predictive models following the flux balance approach is summarized in Fig. 5. After considering nonadjustable constraints, the space of possible steady-state flux distribution is defined by Eqs. 2–4. These constraints represent the invariant structure of the model (the capabilities of cell metabolism). Then, certain flux values, usually substrates uptakes, are incorporated as the inputs of the model (they are adjustable constraints). In order determine the actual flux distribution within the resulting space of feasible distributions, it is assumed that cells have evolved to achieve an optimal behavior owing to evolutionary pressure (*i.e.*, cells regulate its fluxes toward optimal flux distribution). One can, for instance, hypothesize that the metabolic objective of a cell is to maximize its growth, and state this mathematically using a linear objective function to be minimized. Solving the resulting linear programming problem, one obtains the flux distribution that makes the best (optimal) use of the defined metabolic network to satisfy the stated objective function (8, 21).

The aim of flux balance analysis is not to determine the

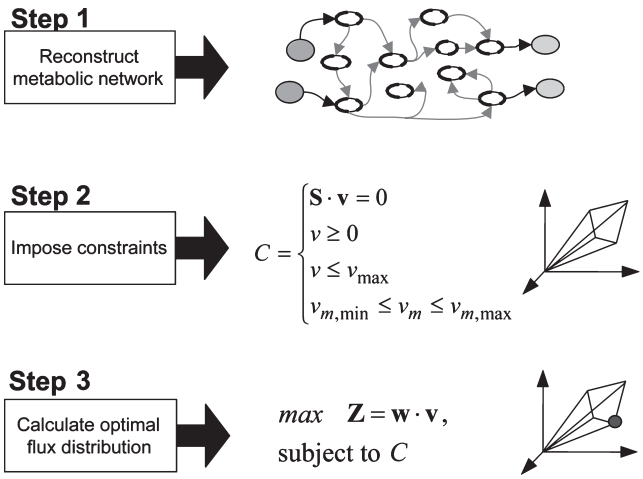

FIG. 5. Procedure to develop a flux balance analysis model. Step 1, Reconstruct the underlying metabolic network. Step 2, Impose the nonadjustable constraints, *e.g.*, toichiometry, irreversibility, capacity, and the adjustable ones, mainly the measured fluxes v_e. Step 3, Determine the actual flux distribution by solving an optimization problem. **Z** denotes the linear objective function and **w** a vector of weights (w_i) on the fluxes **v** used to define the objective function.

flux distribution that corresponds to a set of measurements, but to construct a model capable of predicting the phenotype (flux distribution) that will be expressed under certain conditions (given by the input constraints). Indeed, input constraints do not always correspond to real measurements, as for example, when flux balance analysis is used to investigate certain hypothesis (*e.g.*, can a reduced uptake capacity be the cause of an unexpected cell behavior ?) or to evaluate a range of possibilities (*e.g.*, find the best combination of substrates).

Metabolic objectives and optimization Predictions of flux balance analysis are highly dependent on the objective function being used. To date, the most commonly used objective function has been the maximization of biomass, which leads to predictions consistent with experimental data in 86% of instances for *Escherichia coli* (27) and in 60% for *Helicobacter pylori* (12). Other objective functions have been used, such as minimizing ATP production, minimizing nutrient uptake, or maximizing metabolite production. Moreover, although linear functions are preferred (so as to keep the problem linear) the same philosophy can be applied to other kinds of function. For example, a quadratic function is used by Segre *et al.* (49). The authors tested, with success, that genetically engineered knockout undergo a minimal redistribution with respect to the flux configuration of the wild type cell.

Risks of assumption of optimal behavior The main assumption of flux balance analysis, that is, a cell has evolved to achieve an optimal behavior, has a drawback: the optimal solution may not correspond to the actual flux distribution. To support the assumption of optimal behavior, it must be hypothesized that (i) the cell, forced by evolutionary pressure, has evolved to achieve an optimal behavior with respect to its objective, (ii) we know the objective of the cell, and (iii) the objective can be mathematically stated in a suitable form.

Applications of flux balance analysis In spite of the difficulties mentioned in the previous section, it has been demonstrated that flux balance analysis predictions are consistent with experimental data (12, 27). *E. coli* has been the most well studied microorganism owing to the considerable amount of available data (27, 50). Other models have been developed for *Haemophilus influenzae* (22), *H. pylori* (12), *Saccharomyces cerevisae* (24), *Methanosarcina barkeri* (25), and the human cardiac mitochondria (26). As a result of these efforts, many applications of flux balance analysis have been investigated (1, 8, 21, 48): for example, determination of network properties, support in redundancy studies (*e.g.*, detection of alternate equivalent optima, analysis of the flux variability of a given optimum), interpretation of experimental data (*e.g.*, analysis of robustness against environmental perturbations, qualitative classification of metabolic state based on *in vivo* observations), simulation of genetic modifications (*e.g.*, gene deletion/addition studies, prediction of behavior under knockout conditions, simulation of gradual inhibition), the identification and prioritization of candidate drug targets, the direction of strategies to engineer strains with desired properties, and the analysis of enzyme deficiencies, etc. Furthermore, in recent years, the first medical applications of flux balance analysis have been realized. It is used to investigate the metabolic network of human mitochondria and to evaluate the effect of potential disease treatments (26), and it is applied to optimize the metabolism of cultured hepatocytes used in extracorporeal bioartificial liver devices (43; Yang, H., Yarmush, M. L., Roth, C. M., and Ierapetritou, M., 16th Eur. Symp. Comput. Aided Process Eng., 2006). There is also a reasonable interest on the application of flux balance analysis of whole-plant models. Unfortunately, plant metabolism has certain characteristics: numerous reversible reactions, high compartmentation, non-steady-state conditions, and too many unknown biochemical steps; that make this task very difficult (37, 51).

NETWORK-BASED PATHWAYS ANALYSIS

In network-based pathway analysis, one attempts to elucidate the systemic properties, those that emerge from the entire network as a whole, and capabilities of cell metabolism by analyzing the set of pathways through the cell metabolic network. Linear algebra has been used for this purpose, but this has two limitations: it does not take into consideration the irreversibility constraints, and the linear bases are not an invariant property because they are not unique. For these reasons it has been suggested that the use of convex analysis is more suitable (15).

Thus, taking into consideration the stoichiometric constraints and the irreversibility of certain reactions, a set of possible steady state flux distributions is obtained:

$$\mathbf{P} = \{\mathbf{v} \in \mathbf{R}^{\mathbf{n}} : \mathbf{S} \cdot \mathbf{v} = 0 \text{ and } v_i \geq 0, i \in I\} \quad (6)$$

It is interesting to find an explicit definition of the set of possible steady-state flux distributions (describe the infinite set **P** with a finite set of vectors), because this will allow the interpretation of the overall capabilities of the metabolism of a cell. Interestingly, convex analysis shows that **P** is a convex polyhedral cone that can be spanned by convex combination (non-negative combination) of certain generating vectors (10). Moreover, when all fluxes are irreversible, these generating vectors cannot be decomposed: they encompass as many zeros as possible (52). This means that they represent the most simple steady-state flux distributions.

Elementary modes By extending the simplicity property of the generating vectors to networks with reversible fluxes, the concept of elementary modes was defined (53). Thus, the flux vector ***e*** is an elementary mode (EM) if and only if (54): (i) it satisfies the steady state, (ii) is thermodynamically feasible, and (iii) there is no other non-null flux vector (up to scaling) that both satisfies these constraints and involves a proper subset of its participating reactions.

With this definition, the set of EMs have the following properties (18): (i) There is a unique set of EMs for a given network. Therefore, this set is a systemic property of the metabolism being investigated. Moreover, it is a time-invariant property since only non adjustable constraints are used to define it. (ii) Each EM is genetically independent (cannot be decomposed). Thus, each EM represents a stoichiometrically and thermodynamically feasible route to the conversion of substrates into products, which cannot be decomposed into simpler routes. (iii) EMs are the set of all routes through the network consistent with property ii.

Indeed, EMs span the admissible solution region by convex combination, *i.e.*, each possible steady-state flux distribution **v** can be expressed as a non-negative combination of EMs. If we denote **E** as the matrix formed by placing each EM as a column, and **α** as the vector containing the activity of each EM, then

$$\mathbf{v} = \mathbf{E} \cdot \boldsymbol{\alpha}, \quad \alpha_i \geq 0 \quad (7)$$

Thus, EMs form the set of simplest, systemic routes that can be used to comprehensively describe every valid steady-state flux distribution of the metabolic network. In biological terms, EMs represent all the capabilities of a metabolic network (*i.e.*, all the phenotypes that can be expressed). It is also possible to link the concept of EMs with the role of enzymes: an EM can be considered as the minimal set of enzymes that can operate at steady state.

Extreme pathways The set of extreme pathways (EP) is a subset of elementary modes that satisfies an additional property: no EP can be reconstructed as nonnegative linear combination of other EPs. Therefore, the set of EPs does not include all genetically independent routes, as it is the case for EMs, but just includes a subset still capable of spanning the space P of possible flux distributions by convex combination. An example of the EMs and the EPs of a simple network is shown in Fig. 6. Although both approaches have been successful, the study of system properties using EPs requires careful application (18). The problem arises because EPs do not represent the complete set of simplest (genetically independent) routes within the metabolic network under investigation. On the other hand, it is well known that as the metabolic network under study has more reactions, the number of EMs markedly increases. Although this also occurs in the case of EPs, the problem is not so critical because there are much fewer EPs than EMs.

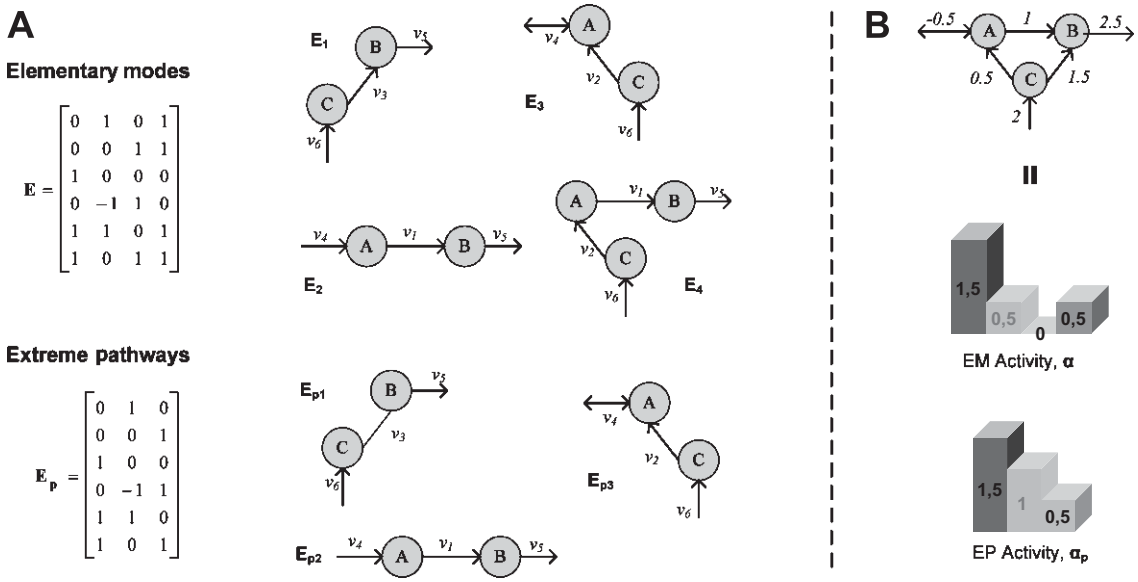

FIG. 6. EMs and EPs. (A) The matrix and the figures represent the set of four elementary modes and the set of three extreme pathways of the network depicted in Fig. 1. Notice that the three extreme pathways are also EMs, and that the fourth EM can be expressed as a combination of the others. (B) Translation of a flux distribution into an EMs/EPs activity pattern. The flux distribution shown in the upper figure can be expressed in terms of flux through EMs or EPs. Note that in the case of the EMs the depicted translation is not the only possible one.

TABLE 2. Applications of elementary modes and extreme pathways analysis

Application	Reference
Elucidation of network properties:	
Identification of pathways	2, 60
Analysis of the importance of reactions	13
Inference of viability of mutants	
Control-effective fluxes	
Detection of reactions correlations (enzyme subsets)	65, 66
Detection of infeasible circles	See Ref. 54
Detection of minimal cut sets	67
Translation of flux distributions into EM/EP patterns	
Particular solution methods	59, 68
α-spectrum	69
Determination of minimal medium requirements	12, 56
Detection of network dead ends (reaction not connected)	12, 56
Analysis of pathway redundancy and robustness	57, 58, 61
Incorporation of information about regulation	11
Support in metabolic engineering:	
Assignment of function to orphan genes	See Ref. 2
Identification of pathways with optimal and suboptimal yields	See Ref. 2
Evaluation of effect of addition/deletion of genes	See Ref. 2
Suggest changes in flux distributions to increase product yield	55
Support in the reconstruction of metabolic maps	70

Part of this information has been extracted from Refs. 2 and 54.

Applications of network-based pathways analysis

Several applications of elementary modes and extreme pathways have been investigated recently (see Table 2). Most of them are found in the context of microbial production, and are used for the study of the metabolism of *E. coli* (45, 55), *H. influenzae* (56, 57), *H. pylori* (58), and *S. cerevisae* (59). However, EMs have also been used in botany. Thus, using EMs analysis, Poolman *et al.* concluded that, in contrast to the traditional view, the Calvin cycle and oxidative pentose phosphate pathway are, in the context of chloroplast, complementary and overlapping components of the same system (60). The structural properties of the plant mitochondrial TCA cycle have also been described using EMs (61). In addition, some medical applications have been reported recently: EMs analysis has been applied to clarifying the metabolism of cultured hepatocytes used in extracorporeal bioartificial liver devices (51), and to investigate the metabolic responses of the rat liver to burn-injury-induced whole-body inflammation (44).

CONCLUSIONS AND PERSPECTIVES

The stoichiometric modelling framework and its principles can be described from a constraint-based perspective. This results in a general definition that is capable of encompassing several related methodologies, which can be classified in two groups: those for system analysis and those for determining the flux solution under certain conditions. Particular attention has been paid to three specific methodologies: metabolic flux analysis, flux balance analysis, and network-based pathway analysis.

Metabolic flux analysis has been commonly used to characterize the exponential growth or steady states in the continuous mode (29–37). However, there is an increasing interest in the application of metabolic flux analysis to monitoring intracellular fluxes over time (38–40). This will be useful for determining the physiological state of cell cultures in industrial environments. In this context, the flux spectrum approach will be useful because it takes into consideration uncertainty of measurements and it enables the estimation of unmeasured fluxes even when there is a lack of measured data (9).

Flux balance analysis is, so far, the only approach applied to developing predictive models in the genome scale (21). Nevertheless, its capabilities are also of great usefulness to deal with simpler networks. Indeed, the cybernetic approach, which is also based on an assumption of optimal behavior, has been used for the dynamic modelling of cells using very simple networks (62). It must also be noted that flux balance analysis is still being improved. In particular, one of the most promising issues is the incorporation of new constraints, such as regulatory, kinetics, or explicit thermodynamics (20).

EMs and EPs are being used to elucidate the systemic properties and capabilities of cell metabolism by analyzing the set of pathways in the cell metabolic network. Despite being recent proposals, both have an increasing number of applications aimed at improving our understanding of biological processes and guiding metabolic engineering. Moreover, new applications are being suggested, such as the use of elementary modes in the development of unstructured, kinetic models compatible with the underlying metabolic network (63, 64; Provost, A., Ph.D. thesis, Université Catholique de Louvain, Louvain-la-Neuve, 2006).

Summarising, stoichiometric modelling is a very active field, which has received much attention recently. Moreover, it is expected that this situation will continue in the foreseeable future. Several methodologies have been developed during the past few years, with an increasing number of applications. Although most of these applications are particularly significant for basic science research, bioprocess industries will also find opportunities to exploit these advances.

ACKNOWLEDGMENTS

This research has been partially supported by the Spanish Government (CICYT-FEDER DPI2005-01180). The first author is recipient of a fellowship from the Spanish Ministry of Education and Science (FPU AP2005-1442).

REFERENCES

1. **Palsson, B. O.:** Systems biology: properties of reconstructed networks. Cambridge University Press New York, New York (2006).
2. **Papin, J. A., Price, N. D., Wiback, S. J., Fell, D. A., and Palsson, B. O.:** Metabolic pathways in the post-genome era. Trends Biochem. Sci., **28**, 250–258 (2003).
3. **Stephanopoulos, G. N., Aristidou, A. A., and Nielsen, J.:** Metabolic engineering: principles and methodologies. Academic Press, San Diego (1998).
4. **Bailey, J. E.:** Mathematical modeling and analysis in biochemical engineering: past accomplishments and future opportunities. Biotechnol. Prog., **14**, 8–20 (1998).
5. **Palsson, B.:** The challenges of in silico biology. Nat. Biotechnol., **18**, 1147–1150 (2000).
6. **Kitano, H.:** Computational systems biology. Nature, **420**, 206–210 (2002).
7. **Wiback, S. J., Famili, I., Greenberg, H. J., and Palsson, B. O.:** Monte Carlo sampling can be used to determine the size and shape of the steady-state flux space. J. Theor. Biol., **228**, 437–447 (2004).
8. **Edwards, J. S., Covert, M., and Palsson, B.:** Metabolic modelling of microbes: the flux-balance approach. Environ. Microbiol., **4**, 133–140 (2002).
9. **Llaneras, F. and Pico, J.:** An interval approach for dealing with flux distributions and elementary modes activity patterns. J. Theor. Biol., **246**, 290–308 (2007).
10. **Rockafellar, R. T.:** Convex analysis. Princeton University Press, Princeton (1996).
11. **Covert, M. W., Schilling, C. H., and Palsson, B.:** Regulation of gene expression in flux balance models of metabolism. J. Theor. Biol., **213**, 73–88 (2001).
12. **Schilling, C. H., Covert, M. W., Famili, I., Church, G. M., Edwards, J. S., and Palsson, B. O.:** Genome-scale metabolic model of *Helicobacter pylori* 26695. J. Bacteriol., **184**, 4582–4593 (2002).
13. **Stelling, J., Klamt, S., Bettenbrock, K., Schuster, S., and Gilles, E. D.:** Metabolic network structure determines key aspects of functionality and regulation. Nature, **420**, 190–193 (2002).
14. **Gombert, A. K. and Nielsen, J.:** Mathematical modelling of metabolism. Curr. Opin. Biotechnol., **11**, 180–186 (2000).
15. **Schilling, C. H., Schuster, S., Palsson, B. O., and Heinrich, R.:** Metabolic pathway analysis: basic concepts and scientific applications in the post-genomic era. Biotechnol. Prog., **15**, 296–303 (1999).
16. **Reder, C.:** Metabolic control theory: a structural approach. J. Theor. Biol., **135**, 175–201 (1988).
17. **Heinrich, R. and Schuster, S.:** The regulation of cellular systems. Chapman & Hall, New York (1996).
18. **Papin, J. A., Stelling, J., Price, N. D., Klamt, S., Schuster, S., and Palsson, B. O.:** Comparison of network-based pathway analysis methods. Trends Biotechnol., **22**, 400–405 (2004).
19. **Llaneras, F. and Pico, J.:** A procedure for the estimation over time of metabolic fluxes in scenarios where measurements are uncertain and/or insufficient. BMC Bioinformatics, **8**, 421 (2007).
20. **Kauffman, K. J., Prakash, P., and Edwards, J. S.:** Advances in flux balance analysis. Curr. Opin. Biotechnol., **14**, 491–496 (2003).
21. **Price, N. D., Papin, J. A., Schilling, C. H., and Palsson, B. O.:** Genome-scale microbial in silico models: the constraints-based approach. Trends Biotechnol., **21**, 162–169 (2003).
22. **Edwards, J. S. and Palsson, B. O.:** Systems properties of the *Haemophilus influenzae* RdMetabolic genotype. J. Biol. Chem., **274**, 17410–17416 (1999).
23. **Edwards, J. S. and Palsson, B. O.:** The *Escherichia coli*

MG1655 in silico metabolic genotype: its definition, characteristics, and capabilities. Proc. Natl. Acad. Sci. USA, **97**, 5528–5533 (2000).
24. **Forster, J., Famili, I., Fu, P., Palsson, B. O., and Nielsen, J.:** Genome-scale reconstruction of the *Saccharomyces cerevisiae* metabolic network. Genome Res., **13**, 244–253 (2003).
25. **Feist, A. M., Scholten, J. C. M., Palsson, B. O., Brockman, F. J., and Ideker, T.:** Modeling methanogenesis with a genome-scale metabolic reconstruction of *Methanosarcina barkeri*. Mol. Syst. Biol., **2**, 2006.0004 (2006).
26. **Thiele, I., Price, N. D., Vo, T. D., and Palsson, B. O.:** Candidate metabolic network states in human mitochondria. Impact of diabetes, ischemia, and diet. J. Biol. Chem., **280**, 11683–11695 (2005).
27. **Edwards, J. S., Ibarra, R. U., and Palsson, B. O.:** In silico predictions of *Escherichia coli* metabolic capabilities are consistent with experimental data. Nat. Biotechnol., **19**, 125–130 (2001).
28. **Klamt, S., Schuster, S., and Gilles, E. D.:** Calculability analysis in underdetermined metabolic networks illustrated by a model of the central metabolism in purple nonsulfur bacteria. Biotechnol. Bioeng., **77**, 734–751 (2002).
29. **Fischer, E., Zamboni, N., and Sauer, U.:** High-throughput metabolic flux analysis based on gas chromatography-mass spectrometry derived 13C constraints. Anal. Biochem., **325**, 308–316 (2004).
30. **Wiechert, W., Mollney, M., Petersen, S., and de Graaf, A. A.:** A universal framework for 13C metabolic flux analysis. Metab. Eng., **3**, 265–283 (2001).
31. **Bonarius, H. P. J., Hatzimanikatis, V., Meesters, K. P. H., de Gooijer, C. D., Schmid, G., and Tramper, J.:** Metabolic flux analysis of hybridoma cells in different culture media using mass balances. Biotechnol. Bioeng., **50**, 299–318 (1996).
32. **Follstad, B. D., Balcarcel, R. R., Stephanopoulos, G., and Wang, D. I.:** Metabolic flux analysis of hybridoma continuous culture steady state multiplicity. Biotechnol. Bioeng., **63**, 675–683 (1999).
33. **Nyberg, G. B., Balcarcel, R. R., Follstad, B. D., Stephanopoulos, G., and Wang, D. I.:** Metabolism of peptide amino acids by Chinese hamster ovary cells grown in a complex medium. Biotechnol. Bioeng., **62**, 324–335 (1999).
34. **Gambhir, A., Korke, R., Lee, J., Fu, P. C., Europa, A., and Hu, W. S.:** Analysis of cellular metabolism of hybridoma cells at distinct physiological states. J. Biosci. Bioeng., **95**, 317–327 (2003).
35. **Schwender, J., Ohlrogge, J., and Shachar-Hill, Y.:** Understanding flux in plant metabolic networks. Curr. Opin. Plant Biol., **7**, 309–317 (2004).
36. **Ratcliffe, R. G. and Shachar-Hill, Y.:** Measuring multiple fluxes through plant metabolic networks. Plant J., **45**, 490–511 (2006).
37. **Lange, B. M.:** Integrative analysis of metabolic networks: from peaks to flux models? Curr. Opin. Plant Biol., **9**, 220–226 (2006).
38. **Herwig, C. and von Stockar, U.:** A small metabolic flux model to identify transient metabolic regulations in *Saccharomyces cerevisiae*. Bioprocess Biosyst. Eng., **24**, 395–403 (2002).
39. **Henry, O., Kamen, A., and Perrier, M.:** Monitoring the physiological state of mammalian cell perfusion processes by on-line estimation of intracellular fluxes. J. Process Control, **17**, 241–251 (2007).
40. **Takiguchi, N., Shimizu, H., and Shioya, S.:** An on-line physiological state recognition system for the lysine fermentation process based on a metabolic reaction model. Biotechnol. Bioeng., **55**, 170–181 (1997).
41. **Teixeira, A. P., Alves, C., Alves, P. M., Carrondo, M. J., and Oliveira, R.:** Hybrid elementary flux analysis/nonparametric modeling: application for bioprocess control. BMC Bioinformatics, **8**, 30 (2007).
42. **Calik, P. and Ozdamar, T. H.:** Metabolic flux analysis for human therapeutic protein productions and hypothesis for new therapeutical strategies in medicine. Biochem. Eng. J., **11**, 49–68 (2002).
43. **Sharma, N. S., Ierapetritou, M. G., and Yarmush, M. L.:** Novel quantitative tools for engineering analysis of hepatocyte cultures in bioartificial liver systems. Biotechnol. Bioeng., **92**, 321–335 (2005).
44. **Nolan, R. P., Fenley, A. P., and Lee, K.:** Identification of distributed metabolic objectives in the hypermetabolic liver by flux and energy balance analysis. Metab. Eng., **8**, 30–45 (2006).
45. **Schmidt, K., Nielsen, J., and Villadsen, J.:** Quantitative analysis of metabolic fluxes in *Escherichia coli*, using two-dimensional NMR spectroscopy and complete isotopomer models. J. Biotechnol., **71**, 175–189 (1999).
46. **Shirai, T., Matsuzaki, K., Kuzumoto, M., Nagahisa, K., Furusawa, C., Shioya, S., and Shimizu, H.:** Precise metabolic flux analysis of coryneform bacteria by gas chromatography–mass spectrometry and verification by nuclear magnetic resonance. J. Biosci. Bioeng., **102**, 413–424 (2006).
47. **Yang, C., Hua, Q., and Shimizu, K.:** Quantitative analysis of intracellular metabolic fluxes using GC-MS and two-dimensional NMR spectroscopy. J. Biosci. Bioeng., **93**, 78–87 (2002).
48. **Shimizu, H.:** Metabolic engineering—integrating methodologies of molecular breeding and bioprocess systems engineering. J. Biosci. Bioeng., **94**, 563–573 (2002).
49. **Segre, D., Vitkup, D., and Church, G. M.:** Analysis of optimality in natural and perturbed metabolic networks. Proc. Natl. Acad. Sci. USA, **99**, 15112–15117 (2002).
50. **Reed, J. L., Vo, T. D., Schilling, C. H., and Palsson, B. O.:** An expanded genome-scale model of *Escherichia coli* K-12 (iJR904 GSM/GPR). Genome Biol., **4**, R54 (2003).
51. **Zhong, J. J.:** Plant cell culture for production of paclitaxel and other taxanes. J. Biosci. Bioeng., **94**, 591–599 (2002).
52. **Pfeiffer, T., Sanchez-Valdenebro, I., Nuno, J. C., Montero, F., and Schuster, S.:** METATOOL: for studying metabolic networks. Bioinformatics, **15**, 251–257 (1999).
53. **Schuster, S., Fell, D. A., and Dandekar, T.:** A general definition of metabolic pathways useful for systematic organization and analysis of complex metabolic networks. Nat. Biotechnol., **18**, 326–332 (2000).
54. **Gagneur, J. and Klamt, S.:** Computation of elementary modes: a unifying framework and the new binary approach. BMC Bioinformatics, **5**, 175 (2004).
55. **Van Dien, S. J., Iwatani, S., Usuda, Y., and Matsui, K.:** Theoretical analysis of amino acid-producing *Escherichia coli* using a stoichiometric model and multivariate linear regression. J. Biosci. Bioeng., **102**, 34–40 (2006).
56. **Schilling, C. H. and Palsson, B. O.:** Assessment of the metabolic capabilities of Haemophilus influenzae Rd through a genome-scale pathway analysis. J. Theor. Biol., **203**, 249–283 (2000).
57. **Papin, J. A., Price, N. D., Edwards, J. S., and Palsson, B. B. O.:** The genome-scale metabolic extreme pathway structure in *Haemophilus influenzae* shows significant network redundancy. J. Theor. Biol., **215**, 67–82 (2002).
58. **Price, N. D., Papin, J. A., and Palsson, B. O.:** Determination of redundancy and systems properties of the metabolic network of *Helicobacter pylori* using genome-scale extreme pathway analysis. Genome Res., **12**, 760–769 (2002).
59. **Schwartz, J. M. and Kanehisa, M.:** Quantitative elementary mode analysis of metabolic pathways: the example of yeast glycolysis. BMC Bioinformatics, **7**, 186 (2006).
60. **Poolman, M. G., Fell, D. A., and Raines, C. A.:** Elementary modes analysis of photosynthate metabolism in the chloroplast stroma. FEBS J., **270**, 430–439 (2003).

61. **Steuer, R., Nesi, A. N., Fernie, A. R., Gross, T., Blasius, B., and Selbig, J.:** From structure to dynamics of metabolic pathways: application to the plant mitochondrial TCA cycle. Bioinformatics, **23**, 1378–1385 (2007).
62. **Ramakrishna, R., Ramkrishna, D., and Konopka, A. E.:** Cybernetic modeling of growth in mixed, substitutable substrate environments: preferential and simultaneous utilization. Biotechnol. Bioeng., **52**, 141–151 (1996).
63. **Provost, A. and Bastin, G.:** Dynamic metabolic modelling under the balanced growth condition. J. Process Control, **14**, 717–728 (2004).
64. **Gao, J., Gorenflo, V. M., Scharer, J. M., and Budman, H. M.:** Dynamic metabolic modeling for a MAB bioprocess. Biotechnol. Prog., **23**, 168–181 (2007).
65. **Schuster, S., Pfeiffer, T., Moldenhauer, F., Koch, I., and Dandekar, T.:** Exploring the pathway structure of metabolism: decomposition into subnetworks and application to *Mycoplasma pneumoniae*. Bioinformatics, **18**, 351–361 (2002).
66. **Cakir, T., Kirdar, B., and Ulgen, K. O.:** Metabolic pathway analysis of yeast strengthens the bridge between transcriptomics and metabolic networks. Biotechnol. Bioeng., **86**, 251–260 (2004).
67. **Klamt, S. and Gilles, E. D.:** Minimal cut sets in biochemical reaction networks. Bioinformatics, **20**, 226–234 (2004).
68. **Schwarz, R., Musch, P., von Kamp, A., Engels, B., Schirmer, H., Schuster, S., and Dandekar, T.:** YANA—a software tool for analyzing flux modes, gene-expression and enzyme activities. BMC Bioinformatics, **6**, 135 (2005).
69. **Wiback, S. J., Mahadevan, R., and Palsson, B. O.:** Reconstructing metabolic flux vectors from extreme pathways: defining the alpha-spectrum. J. Theor. Biol., **224**, 313–324 (2003).
70. **Cornish-Bowden, A. and Cardenas, M. L.:** From genome to cellular phenotype—a role for metabolic flux analysis? Nat. Biotechnol., **18**, 267–268 (2000).